

Andrew Park

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EDUCATION

Duke University, Trinity College of Arts & Sciences

B.S. Mathematics, B.S. Computer Science

Durham, NC

Expected May 2028

- GPA 3.96/4.00. Julia Dale Prize for first-year mathematical excellence; Dean's List with Distinction.
- Graduate mathematics:** Mathematical Finance, Financial Derivatives, Real Analysis, Algebraic Structures, Measure & Integration*, Numerical Linear Algebra*. **Probability & CS:** Advanced Probability, Data Structures & Algorithms, Computer Architecture, Graphics Software Architecture, Discrete Mathematics & Proofs*.

*in progress, Fall 2026

Stuyvesant High School: ACT 36; National Merit Scholar

New York, NY

EXPERIENCE

Quantitative Research Intern, DataBallR

May 2026 – Present

Daily RAPM, a public multi-league basketball valuation model at andrewjparkus.github.io/model.html

- Showed that a weighted blend of the **Las Vegas closing line** and my player ratings predicts NBA game margins better than the closing line alone. Weights were fit on earlier seasons only and scored on each held-out season across **12,034 games and 12 seasons**; the fit put roughly **30% of its weight on my number** every time.
- Built the 2026-27 season projection end to end and published it before the league released its schedule: a minutes and games played model, an automated depth chart and rotation builder, per-player wins contributed, and a Monte Carlo simulation producing standings, playoff and title odds, with all 30 team win totals frozen in public version control.
- Directed the underlying daily valuation system, which rebuilds every rating from scratch on *every* game day: a regularized regression over **49M+ play-by-play possessions** placing **10,681** NBA, NCAA Division I and G League players on one scale, benchmarked publicly against EPM, BPM, RAPTOR and DARKO.

PRUV Research Fellow, Duke Mathematics

April 2025 – Present

Senior thesis under Professor J. Thomas Beale

Durham, NC

- Model atmospheric flow on brown dwarfs, which spin fast enough to be visibly flattened. The literature solves these atmospheres on a sphere; I solve the deformed figure instead and measure what that assumption costs, finding a rotational signal thousands of times stronger than the spherical case predicts.
- Implemented Professor Beale's method for solving partial differential equations on curved surfaces as a 4,500-line Julia library, then made it tractable at scale with sparse, matrix-free numerical linear algebra. The surface never has to be meshed.
- Verified it against exact solutions on **10 distorted planetary shapes**, confirming its predicted second-order accuracy on every one; results are being written up as a 40-page manuscript.

Teaching Assistant & Grader, Duke Mathematics

July 2025 – Present

- Teaching assistant for **graduate Mathematical Finance as an undergraduate**, plus proof-based problem sets for 400+ students in Linear Algebra, Multivariable Calculus and Differential Equations.

PROJECTS

Election Forecasting Engine & Prediction-Market Portfolio

September 2021 – Present

- Built a U.S. election forecasting engine at county and precinct level on returns dating to 1912. Rather than scoring a county on a single partisan axis, I fit an **ideology vector** for each and used the correlations between them to push national, state and local swings into races with no polling of their own. Directed seven contributors.
- Ran the portfolio on **Polymarket and Kalshi** across four cycles (2022, 2024, 2025 local races, 2026 ongoing): took positions wherever the model's probability diverged from market price enough to be positive expected value, sized by **fractional Kelly**, then **hedged across presidential, gubernatorial, Senate and House** contracts so a national or state polling miss could not run through the whole book.

Weather Derivatives Pricing Engine

2026

- CME weather contracts settle on a handful of major-city stations, so a grower or utility a hundred miles out hedges with weather that is not its own. Built a pricing engine that closes that gap: it fits 75 years of daily temperature for **all 3,107 U.S. counties**, models how each reverts toward its own seasonal normal, and prices degree day options county by county, validated on held-out years against quoted prices.

Duke Poker Club, Founder & President

January 2025 – Present

- Re-launched the club to 100+ members, secured a collegiate Kalshi partnership, trained the competitive tournament team, and was interviewed by *The Wall Street Journal* on college poker and prediction markets.

ADDITIONAL INFORMATION

Languages: Python, C++, Julia, SQL, Java, MIPS

Tools: NumPy, pandas, scikit-learn, XGBoost, Arrow/Parquet, Git, $\text{\LaTeX}$

Methods: statistical modeling, stochastic processes, Monte Carlo, backtesting, numerical methods, portfolio construction and sizing

Spoken: English and Korean (native), French (fluent)

Interests: classical piano; violin on a 2023 Grammy-winning orchestral recording; nineteenth-century literature